Julián Szereszewski

Buenos Aires, Argentina
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Education

University of Buenos Aires March 2019 – April 2026
Licentiate in **Physics** (B.Sc. + M.Sc. equivalent), GPA 9.3 / 10. *FCEyN — UBA*

National College of Buenos Aires March 2014 – December 2018
High School diploma. *CNBA — UBA*

Research Experience

Supervised Program for Alignment Research (SPAR) February 2026 - May 2026
Investigating a geometry based explanation of *subliminal learning* using steering vectors extracted as the difference between biased and unbiased activations. *Advisor: Yuxiao Li*

Dynamical Systems Lab (LSD) February 2025 – December 2025
Worked on Kolmogorov-Arnold Networks (KANs) for my thesis project to model dynamical systems as a more interpretable alternative to traditional Neural Networks. *Advisor: Gabriel B. Mindlin*

Work Experience

Performance Analyst — Avature August 2024 – Present
I develop and maintain statistical analysis tools in Python to gain insights from web server response times. *Semi-Senior*

QA Engineer — Avature July 2021 – August 2024
I analyzed database inconsistencies with SQL and developed automated tests using Selenium, JavaScript and PHP. *Semi-Senior*

Relevant Coursework

- Statistical Mechanics
- Machine Learning
- Modeling of Complex Systems
- Statistics for Experimental Physics
- Quantum Mechanics
- Linear Algebra
- Classical Mechanics
- Differential and Integral Calculus
- Complex Analysis

Courses & Conferences

Effective Altruism Global (EAG) — London, UK May 2026

Technical AI Safety Course — BlueDot Impact January 2026

Deep Learning Spring School — FCEyN October 2025

Generative Image Models Based on Deep Neural Networks — FCEyN August 2025

Scientific Symposium on AI and Applications — UdeSA November 2024 & September 2025

Papers

J. Szereszewski, F. Fainstein, L. E. Fernandez, G. B. Mindlin. (2026) *Inferring hidden forcing in a biological oscillator using Kolmogorov–Arnold networks*. arXiv:2606.08479.

Conference Presentations

J. Szereszewski, N. Waehner, M. Smith. *Sparse Autoencoders for Interpretable Memory Retrieval in LLMs*. Poster presented at the Y-Hat Hackathon, FCEyN (December 2025). Buenos Aires, Argentina.

J. Szereszewski, F. Fainstein, L. E. Fernandez, G. B. Mindlin. *Kolmogorov Arnold Networks for the reconstruction of dynamical systems from data*. Poster presented at the Scientific Artificial Intelligence and Applications Symposium (SCIAA), UdeSA (September 2025); and the Deep Learning Spring School, FCEyN (October 2025). Buenos Aires, Argentina.

J. Szereszewski, F. Otero Zappa, A. Kleiman, M. Zanini, D. Grondona. *Design and Assembly of a Scalable Plasma Reactor for Water Remediation*. Poster presented at the Annual Reunion of Argentinian Physics Asociation (RAFA), UNSL (September 2024). San Luis, Argentina.

Achievements

Hackathon Winner for the poster presented at the NLP track of the Y-Hat Hackathon (FCEyN-UBA). *December 2025, Buenos Aires, Argentina.*

Best Poster Award for the poster presented at the Deep Learning School (FCEyN-UBA). *October 2025, Buenos Aires, Argentina.*

Finalist at the National Stage of the Argentine Mathematical Olympiad (OMA). *November 2018, La Falda, Córdoba, Argentina.*

Nominated for the Metropolitan Argentine Mathematical Olympiad (OMA). *August 2018, Mar del Plata, Buenos Aires, Argentina.*